INTELLIGENT FREIGHT FORECASTING & VESSEL CHARTER OPTIMIZATION

ML + OPTIMIZATION ARCHITECTURE

Problem Statement: Ministry of Steel / SAIL (Steel Authority of India Limited) • **Corridor:** Overseas Raw Materials → Indian East Coast Ports (Haldia, Paradip, Vizag, Dhamra) • Minimizing Landed Cost ($/MT) Under Draft, Demurrage & Stockout Constraints

P10 / P50 / P90	MILP Solvers	7d to 90d	6 East Coast Ports
PROBABILISTIC QUANTILES	PULP / OR-TOOLS / HIGHS	PLANNING HORIZONS	TIDAL & DRAFT MODELING

0 CONSIGNMENT & REQUISITION

Cargo Requirements
INPUT SPEC

- **Commodity:** Coking Coal / Thermal Coal / Met Coke / Limestone
- **Consignment Quantity:** `Q = 75,000 MT` (±10% MOLOO)
- **Origin Terminal:** Taboneo (IDN), Gladstone (AUS), Norfolk (USA), Maputo (MOZ)
- **Discharge Port:** Haldia Dock Complex (HDC) / Paradip / Visakhapatnam
- **Laycan Window:** `[T_start, T_end]` (e.g., Oct 15 - Oct 25)
- **Plant Stockout Deadline:** `T_crit` (Critical inventory run-out at Bokaro/Rourkela)

```
Required Arrival: t_arrival ≤ T_crit
Laycan Window: t_dep + t_voyage ∈ [T_start, T_end]
```

Current Baseline Problem
VULNERABILITY

- **Reactive Spot Fixing:** Chartering day-to-day without forward forecasting leads to locking vessels during market spikes.
- **Draft Restrictions:** Haldia riverine draft (<8.5m) forces dead-freight or costly unexpected emergency lightering.
- **Demurrage Bleed:** Berth queues at Paradip & Haldia cause $20,000/day bleed on unoptimized laytime terms.
- **Sub-optimal Vessel:** Booking Handysize when Panamax + lightering saves $4.50/MT landed.

Vessel Sizing & Scale Curve
ECONOMICS

CLASS	CAPACITY	BASE SPOT $/MT	DRAFT REQ
Handysize	28k - 38k MT	$22.50 / MT	~9.8m
Supramax	50k - 58k MT	$18.80 / MT	~12.2m
Panamax	70k - 82k MT	$14.20 / MT	~14.5m
Capesize	150k - 180k MT	$11.80 / MT	~18.2m

Contract Modalities
CHARTER MODES

MODE	BASIS	EXPOSURE
Voyage Charter	$/MT lump sum	Fuel included; demurrage exposed
Time Charter (TC)	$/day hire	Full bunker + canal + port dues risk
COA Contract	Multi-shipment	Quarterly volume commitments

Engine Execution Triggers
AUTOMATION

- **Scheduled Run:** Daily automated optimization at 06:00 UTC
- **Market Event:** Trigger on Baltic Index daily swing > 4.5%
- **Interactive Requisition:** Real-time simulation on user query

1 MULTI-SOURCE INGESTION & LAKE

Baltic Exchange Indices
DAILY TS

- **BDI:** Baltic Dry Composite Benchmark
- **BCI:** Baltic Capesize Index (180,000 DWT)
- **BPI:** Baltic Panamax Index (82,500 DWT)
- **BSI:** Baltic Supramax Index (58,328 DWT)
- **BHSI:** Baltic Handysize Index (38,200 DWT)

```
Features: 7d/14d/30d Rolling Mean, Volatility σ,
RSI(14), MACD(12,26,9), Cross-Index Spreads
```

Bunker Fuel Benchmarks
ENERGY FEED

- **VLSFO (0.5% S):** Singapore & Fujairah Hubs
- **LSMGO (0.1% S):** Marine Gas Oil ECA compliant
- **Brent Crude:** Macro energy momentum indicator
- **Hi-5 Spread:** VLSFO vs HSFO scrubber arbitrage

```
BAF Adjustment = Consumption_Rate × ΔP_bunker
```

Port Master & Constraints
STATIC/TIDAL

PORT	MAX DRAFT	MAX LOA	DISCHARGE RATE
Haldia (HDC)	7.5m - 8.5m	195m	12,000 MT/day
Paradip	14.5m - 16.0m	260m	25,000 MT/day
Vizag / Gang.	16.5m - 18.5m	300m	35,000 MT/day
Dhamra	18.0m	320m	30,000 MT/day
Sandheads (Anch)	Deepwater	Cape	15,000 MT/day (FSU)

Vessel Fleet Specifications
FLEET DB

- **Handysize:** 28k-39k DWT | Draft: 10.0m | LOA: 180m
- **Supramax/Ultra:** 50k-64k DWT | Draft: 12.8m | LOA: 199m
- **Panamax/Kamsar:** 65k-85k DWT | Draft: 14.5m | LOA: 229m
- **Capesize:** 120k-200k DWT | Draft: 18.2m | LOA: 292m
- **Speed/Consumption:** Ballast 13.0 kts, Laden 12.0 kts

AIS Congestion & Telemetry
REAL-TIME

- **Port Anchorage Queue:** Real-time vessel count at roadstead
- **Historical Waiting Time:** Moving median wait (days)
- **Macro & FX:** USD/INR spot + forward, China Steel PMI
- **Weather / Monsoon:** SW monsoon (Jun-Sep), cyclones

2 ML FREIGHT RATE FORECASTING

Feature Engineering Pipeline
PREPROCESSING

- **Temporal Lags:** Lags t-1, t-3, t-7, t-14, t-30 of Route Spot Rates
- **Exogenous Regressors:** BDI, BPI, BSI, VLSFO Singapore, Coal Argus Index
- **Calendar Cyclics:** sin(2πd/365), cos(2πd/365), Monsoon indicator
- **Cross-Feature Ratios:** Route Rate / BDI, Bunker / Freight Intensity
- **Target Horizons:** $h \in \{7, 15, 30, 60, 90\}$ days ahead

```
X_t = [F_{t-k}, BDI_{t-k}, Bunker_{t-k}, Macro_t, Season_t]
```

Ensemble Quantile Models
CORE ML

- **LightGBM Quantile Regressors:** Pinball Loss optimization for $\alpha \in \{0.10, 0.50, 0.90\}$
- **Prophet Bayesian Time Series:** Additive seasonality + changepoint trend detection
- **Temporal Fusion Transformer (TFT):** Multi-horizon self-attention networks
- **Validation:** Purged Walk-Forward Expanding Window CV (zero data leakage)

```
Pinball Loss:
L_q(y, y_pred) = max(q(y - y_pred), (1 - q)(y_pred - y))
```

Forecast Confidence Envelopes
RATE PREDICTIONS

METRIC	PERCENTILE	OPERATIONAL MEANING
P10	10th %ile	Optimistic / Trough Rate
P50	50th %ile	Base Expected Rate
P90	90th %ile	Pessimistic / Spike Rate

```
Spread Δ = P90 - P10 indicates Market Volatility Risk
```

Backtest Validation Metrics
ACCURACY

HORIZON	MAE ($/MT)	MAPE (%)	DIR ACCURACY
7 Days	$0.48 / MT	3.2%	86.4%
30 Days	$0.95 / MT	6.8%	81.2%
90 Days	$1.82 / MT	12.4%	74.5%

Explainable AI (XAI)
SHAP ATTRIB

- **VLSFO Price Shock:** Contributes +38% to freight upside
- **BPI Momentum:** Contributes +25% trend alignment
- **Monsoon Seasonality:** Dampens port handling throughput
- **China Steel Production:** Macro demand pull signal

3 PORT & VESSEL FEASIBILITY

Physical Constraint Engine
RULE MATRIX

- **Draft Verification:** Vessel Laden Draft ≤ Port Permissible Draft (LAT + Tide)
- **LOA (Length Overall):** LOA_vessel ≤ Max Permitted Berth LOA
- **Beam Limitation:** Lock gates and gantry crane outreach limits
- **Air Draft:** Conveyor bridge clearances at riverine berths

```
IsDirectFeasible = (Draft_v ≤ Draft_port) ∧
(LOA_v ≤ MaxLOA_port) ∧ (Beam_v ≤ MaxBeam_port)
```

Haldia Bottleneck: Sandheads Logic
CORE SIH FOCUS

Haldia Dock Complex cannot handle deep-draft Panamax/Capesize (>8.5m). Two routing pathways exist:

- **Pathway A (Direct Smaller Vessel):** Charter Handysize / Supramax (shortage in supply, higher $/MT freight).
- **Pathway B (Mother-Daughter Lightering):** Charter large Panamax/Capesize (cheap freight) → lighten at Sandheads / Sagar → barge to Haldia.

```
If Draft_v > 8.5m & Port == HDC:
⇒ Trigger Forced Lightering Sub-Model (Sandheads)
```

Sandheads Transshipment Specs
OPERATIONS

- **Lightering Anchorage:** Sandheads deep roadstead (20m water)
- **Floating Crane Transshipment:** Rate: 15,000 MT/day
- **Weather Cutoff:** Wave swell > 2.2m halts operations
- **Daughter Barge Transit:** 18h transit time from Sandheads to HDC

Feasibility Routing Matrix
ROUTE LOOKUP

VESSEL CLASS	HALDIA	PARADIP	VIZAG
Handysize (35k)	Direct BERTH	Direct	Direct
Supramax (58k)	Partial Draft	Direct	Direct
Panamax (75k)	LIGHTERING REQ	Direct	Direct
Capesize (150k)	LIGHTERING REQ	Direct	Direct

Dynamic Tidal Windows
HYDRODYNAMICS

- Spring Tide vs Neap Tide water level variations (+1.2m)
- Hooghly river siltation seasonal dredging notices
- Under-Keel Clearance (UKC) safety buffer: 0.9m minimum

4 MARITIME RISK & DEMURRAGE

Port Congestion & Queue
QUEUE THEORY

- **M/M/c Berth Queue Model:** Arrival rate λ vs service rate μ
- **Expected Waiting Days:** W_q = f(AIS vessels waiting, berth availability)
- **Paradip / Haldia Average:** 2.5 to 6.0 days waiting time during peak coal intake

```
Turnaround = W_queue + (Cargo_MT / Discharge_Rate)
```

Demurrage Exposure Engine
COST RISK

- **Laytime Allowed:** Based on agreed Charter Party (CP) discharge rate
- **Demurrage Rate:** $18,000 to $32,000 per day (vessel size dependent)
- **Despatch Bonus:** Half demurrage earned if discharged early

```
Allowed Days = Cargo_MT / CP_Discharge_Rate
Overstay Days = max(0, Turnaround - Allowed Days)
Demurrage Exposure = Overstay Days × Demurrage_Rate
```

Disruption Warnings
EARLY ALERTS

- **Cyclone Risk:** Bay of Bengal alert during Oct-Nov & Apr-May
- **Draft Derating:** Monsoon siltation reduces permissible draft by 0.4m
- **Canal Delays:** Malacca Strait congestion & pilot availability

Terminal Congestion Benchmarks
AIS TELEMETRY

TERMINAL	MEAN WAIT	CONGESTION INDEX	STATUS
Haldia HDC	4.8 Days	High (0.82)	Bottleneck
Paradip Mech.	3.2 Days	Moderate (0.58)	Normal
Vizag VCT	1.5 Days	Low (0.24)	Fluid

Idle / Repositioning Engine
ASSET EFFICIENCY

- **Empty Ballast Penalty:** Returning empty to Australia adds $/day cost
- **Backhaul Matching:** Matching iron ore export from Paradip → China to offset deadheading cost
- **Triangulation Routing:** AUS → India → S.Africa → India

5 LANDED COST & MILP ENGINE

Total Landed Cost Formula
OBJECTIVE COMPONENT

```
Landed Cost ($/MT) = [
  Base Freight (Forecast P50 × Q)
  + Bunker Adjustment Factor (BAF)
  + Demurrage Exposure ($/day × Wait_days)
  + Lightering & Barging Cost ($/MT × Q_lightened)
  + Port Dues, Pilotage & Handling Charges
] / Cargo Volume (Q)
```

- **Lightering Fee:** ~$7.50/MT for transshipment at Sandheads
- **Port Charges:** $2.20/MT standard East Coast tariff

MILP Mathematical Formulation
SOLVER SPEC

- **Decision Variable:** `x_{v,r,t} ∈ {0, 1}` (Vessel v, Route r, Charter Day t)
- **Timing Variable:** `y_fix_now ∈ {0, 1}` vs `y_hold ∈ {0, 1}`

```
min Σ_{v,r,t} x_{v,r,t} × [ LandedCost(v,r,t) + λ × Risk(v,r,t) ]

Subject to:
• Σ_{v,r,t} x_{v,r,t} × Capacity_v ≥ Cargo_Required
• t + VoyageTime(v,r) ≤ Plant_Deadline (T_crit)
• t ∈ Laycan_Window
• Draft_v ≤ PortDraft_r + M × z_{lightering}
• Σ_{v,r,t} x_{v,r,t} = 1 (Single shipment assignment)
```

Option Trade-Off Evaluation
WHAT-IF SIM

OPTION	FREIGHT	LIGHTERING	DEMURRAGE	TOTAL $/MT
1. Supramax Direct	$22.50	$0.00	$1.80	**$24.30**
2. Panamax + Lighten	$14.20	$4.80	$1.20	**$20.20 (Best)**
3. Cape to Paradip	$11.80	$0.00 + Rake	$2.40	**$21.90**

Monte Carlo Risk Simulation
10,000 RUNS

- **Value at Risk (VaR 95%):** Maximum landed cost $24.70/MT
- **Conditional VaR (CVaR):** Expected cost in worst 5% = $26.10/MT
- **Demurrage Volatility:** Modeled via log-normal distribution

Charter Timing Logic
DECISION RULE

```
If E[Rate_{t+k}] > Rate_today + Holding_Cost:
⇒ Recommendation = "FIX NOW" (Protect against rally)
Else If E[Rate_{t+k}] < Rate_today - Volatility_Buffer:
⇒ Recommendation = "WAIT / HOLD" (Harvest softening)
```

6 DECISION OUTPUT & RECOMMENDATION

ACTION: FIX NOW (Confidence 88%)

`Projected Savings: $307,500 ($4.10/MT)`

OPTIMAL VESSEL CLASS
Kamsarmax / Panamax (75k-82k DWT)

OPTIMAL ROUTING STRATEGY
Sandheads Lightering → Haldia Dock

METRIC	P10 (LOW)	P50 (EXPECTED)	P90 (HIGH)
Forecast Ocean Freight ($/MT)	$13.40	**$14.20**	$16.80
Lightering & Port Dues ($/MT)	$4.60	**$4.80**	$5.10
Demurrage Exposure ($/MT)	$0.70	**$1.20**	$2.80
TOTAL LANDED COST ($/MT)	**$18.70**	**$20.20**	**$24.70**

Decision Rationale: BPI and VLSFO forward curves project an 8.4% freight rally over the next 14 days due to Chinese steel restocking. Fixing a Panamax today at $14.20/MT + Sandheads lightering ($4.80) locks in $20.20/MT landed cost, outperforming both Supramax direct ($24.30) and delayed spot chartering ($22.90 expected in 10 days).

Active Maritime Risk Warnings
ACTION REQUIRED

- **Haldia River Draft Siltation:** High tide window required for 8.2m daughter barges. Pilot booking required 48h in advance.
- **Demurrage Exposure Warning:** Paradip coal berths reporting 4 vessels in roadstead. Recommend securing 7 days laytime in Charter Party.
- **Ballast Employment:** Target Paradip pellet export backhaul to Qingdao to reduce voyage charter premium by $1.80/MT.

Hinterland Rail & Rake Linkage
SAIL STEEL PLANTS

PLANT DESTINATION	OPTIMAL PORT	RAIL FREIGHT	RAKE TURNAROUND
Durgapur (DSP)	Haldia (HDC)	$4.20 / MT	1.8 Days (BOXN)
Bokaro (BSL)	Haldia / Paradip	$5.80 / MT	2.4 Days
Rourkela (RSP)	Paradip / Dhamra	$4.90 / MT	2.0 Days

Production Tech Stack Blueprint
IMPLEMENTATION

- **ML Pipeline:** LightGBM, Prophet, PyTorch (TFT), scikit-learn, MLflow
- **Optimization:** PuLP, SciPy, Google OR-Tools, HiGHS MILP
- **Backend API:** FastAPI (Python), Celery Tasks, Redis Cache, PostgreSQL
- **Frontend UI:** React, TailwindCSS, Chart.js, Leaflet AIS Map